Exercise Solution for Chapter-03: Medium Access Control

1. **What is the main physical reason for the failure of many MAC schemes known from wired networks? What is done in wired networks to avoid this effect?**

   In wired networks, all stations can hear each other because the signal strength stays strong along the wire. So, when one station transmits, others can detect it — like in Ethernet (10Base2), which uses CSMA/CD.

   But in wireless networks, stations often can't hear each other, even though they can talk to a central station. That's why carrier sensing (CS) doesn't work well — it happens at the sender, while collisions happen at the receiver.

   In wired networks, this isn't a problem, but in wireless networks, CS and collision detection at the sender often fail because the sender may not detect other signals or collisions.

2. **Recall the problem of hidden and exposed terminals. What happens in the case of such terminals if Aloha, slotted Aloha, reservation Aloha, or MACA is used?**

   In Aloha, stations just send data without checking the medium, so no carrier sensing is done. This means hidden stations can cause collisions, and exposed terminals are not an issue.
   The same applies to Slotted Aloha**,** except access happens in time slots.

   Reservation schemes need a central station that all others can hear. If that's missing, the system fails. So, no hidden or exposed terminal problem here.

   MACA is designed to handle hidden and exposed terminals in wireless networks without a central station. But it can still fail if:

   - Communication is asymmetric, or
   - The network changes rapidly (e.g., moving stations enter collision range).

3. **How does the near/far effect influence TDMA systems? What happens in CDMA systems? What are countermeasures in TDMA systems, what about CDMA systems?**

   In TDMA systems, it doesn't matter if terminals are near or far — as long as the signal arrives on time in the correct time slot. Terminals adjust their transmission power and send a bit early, depending on their distance to the receiver.

   In CDMA systems, terminals must frequently adjust power (like 1500 times per second in UMTS) so that all signals reach the base station with similar strength. If not, stronger signals can block weaker ones, since they're not separated by time.

4. **Who performs the MAC algorithm for SDMA? What could be possible roles of mobile stations, base stations, and planning from the network provider?**

   SDMA is usually managed by the network provider, who plans base station locations based on area, capacity, and topology. When the system runs, base stations help assign terminals based on signal strength or network load.

   Mobile terminals assist by sending info about the signal strength they receive and can also request to switch access points.

5. **What is the basic prerequisite for applying FDMA? How does this factor increase complexity compared to TDMA systems? How is MAC distributed if we consider the whole frequency space as presented in chapter 1?**

Modulation shifts baseband signals to a carrier frequency, which needs analog components. Receivers use filters to pick up signals at specific frequencies, and different antennas may be needed based on the frequency.

- In TDMA, all devices use one frequency, so receivers wait on that frequency.
- In FDMA, receivers must scan different frequencies to find signals.

MAC is handled at multiple levels:

- WRC (World Radio Conferences) assign global frequencies (e.g., 2 GHz for IMT-2000).
- ITU manages global use, while national authorities handle local regulations.
- Network operators manage frequency use based on planning and load.
- Base stations assign frequencies to terminals as needed.
- In WLANs, network admins assign frequencies to create cells.

6. **Considering duplex channels, what are alternatives for implementation in wireless networks? What about typical wired networks?**

Wireless networks use different frequencies, time slots, or codes to create duplex channels (for two-way communication). In contrast, wired networks usually use separate wires for sending and receiving. More advanced methods like echo cancellation can also be used.

7.

i. **What are the advantages of a fixed TDM pattern compared to random, demand-driven TDM?**
ii. **Compare the efficiency in the case of several connections with fixed data rates or in the case of varying data rates.**
iii. **Now explain why traditional mobile phone systems use fixed patterns, while computer networks generally use random patterns.**
iv. **In the future, the main data being transmitted will be computer-generated data. How will this fact change mobile phone systems?**

With fixed TDM patterns, terminals are simple — they just need to stay synchronized to receive data. This is used in classical telecom systems like ISDN and SDH.

In contrast, networks like Ethernet, the Internet, and WLANs are demand-driven, meaning there's no need to reserve time slots before sending data — making them more flexible.

Today, data traffic has grown much more than voice. So, networks must be optimized for data.

- WLANs were designed for data, but audio (like voice calls) can still be tricky.
- Mobile phone systems were built mainly for voice and use circuit switching, but now they are evolving:
  - GPRS adds data support to GSM
  - UMTS uses IP in the core network to handle more data efficiently.

8. Explain the term interference in the space, time, frequency, and code domain. What are countermeasures in SDMA, TDMA, FDMA, and CDMA systems?

Interference and Countermeasures:

- **SDMA:** Interference occurs when senders are too close.
  - ✓ Maintain minimum distance between terminals or base stations.
- **TDMA:** Interference happens when senders transmit at the same time.
  - ✓ Use tight synchronization and guard times between slots.
- **FDMA:** Interference occurs if senders use the same frequency.
  - ✓ Assign different frequencies, use guard bands, or apply frequency hopping.
- **CDMA:** Interference arises from non-orthogonal codes (overlapping signals).
  - ✓ Use orthogonal or quasi-orthogonal codes to separate transmissions.

9. Assume all stations can hear all other stations. One station wants to transmit and senses the carrier idle. Why can a collision still occur after the start of transmission?

Even in a vacuum, radio waves travel at the speed of light, and they go slower through matter. So, a sender may sense the medium as idle and start transmitting — but before the signal reaches another sender, that sender also senses idle and starts sending. This causes a collision, which is why collision detection (CD) is used in classic CSMA/CD Ethernet networks.

10. What are benefits of reservation schemes? How are collisions avoided during data transmission, why is the probability of collisions lower compared to classical Aloha? What are disadvantages of reservation schemes?

Once a reservation is made, no more collisions occur (if there are no errors). Reservation schemes can guarantee bandwidth, delay, and low jitter, so transmission is smooth. Compared to classical Aloha, the chance of collision is lower because the contention phase is short, and most time is used for collision-free transmission. The downside is latency — terminals must reserve the medium first, which wastes time when the network is lightly loaded.

11. How can MACA still fail in case of hidden/exposed terminals? Think of mobile stations and changing transmission characteristics.

In asymmetric conditions like the hidden terminal scenario, MACA can fail.

For example, if station C transmits with high power but can't hear station B, it won't receive B's CTS (Clear to Send). So, C starts sending and causes a collision at B, even though B tried to reserve the channel.

12. Which of the MAC schemes can give hard guarantees related to bandwidth and access delay?

- Fixed TDMA schemes can give hard guarantees (e.g., in ISDN, SDH, GSM), which is why they're used in classical phone systems.
- Implicit reservations can also give guarantees after the reservation is made.
- Any centralized system (with a base station or access point) can provide such guarantees.
- But non-deterministic schemes like CSMA/CA and MACA cannot offer hard guarantees

13. How are guard spaces realized between users in CDMA?

The guard space between users in CDMA systems is the orthogonality between the spreading codes. The lower the correlation is, the better is the user separation.

14. Redo the simple CDMA example of section 3.5, but now add random 'noise' to the transmitted signal (–2,0,0,–2,+2,0). Add, for example, (1,–1,0,1,0,–1). In this case, what can the receiver detect for sender A and B respectively? Now include the near/far problem. How does this complicate the situation? What would be possible countermeasures?

Step 1: Two signals are sent by two users, A and B.

- User A sends: (-2, 0, 0, -2, +2, 0)
- User B sends: (+1, -1, 0, +1, 0, -1)

Step 2: The combined transmitted signal is the sum of A and B's signals:

(-2 + 1, 0 - 1, 0 + 0, -2 + 1, +2 + 0, 0 - 1) = (-1, -1, 0, -1, +2, -1)

Step 3: The receiver tries to decode signal A by multiplying the received signal with A's spreading code:

A's spreading code is (-1, +1, -1, +1, +1, +1)

Calculate:

(-1)*(-1)* + *(-1)*(+1) + 0*(-1) + (-1)*(+1)* + *(+2)*(+1) + (-1)*(+1) = 1 - 1 + 0 - 1 + 2 - 1 = 0

Step 4: The receiver tries to decode signal B by multiplying the received signal with B's spreading code:

B's spreading code is (+1, +1, -1, +1, -1, +1)

Calculate:

(-1)*(+1)* + *(-1)*(+1) + 0*(-1) + (-1)*(+1)* + *(+2)*(-1) + (-1)*(+1) = -1 -1 + 0 -1 -2 -1 = -6

Step 5: The receiver can decide more easily that B sent binary 0 because the result is -6 (a strong negative number), while A's result is 0, which is less clear.

Step 6: Now, imagine noise and a "near/far" problem where B's signal and noise are amplified by 20 times:

- Signal A: (-1, +1, -1, -1, +1, +1)
- Signal B multiplied by 20: (-20, -20, +20, -20, +20, -20)
- Noise multiplied by 20: (+20, +20, 0, +20, 0, -20)

Step 7: Add these three signals to get the new received signal:

(-1 - 20 + 20, +1 - 20 + 20, -1 + 20 + 0, -1 - 20 + 20, +1 + 20 + 0, +1 - 20 - 20) = (-1, -39, 19, -1, 21, -39)

Step 8: The receiver decodes again by multiplying the new signal with A's spreading code:

(-1)*(-1)* + *(-39)*(+1) + 19*(-1) + (-1)*(-1)* + *21*(+1) + (-39)*(+1) = 1 - 39 - 19 + 1 + 21 - 39 = -74

Step 9: The receiver decodes again by multiplying the new signal with B's spreading code:

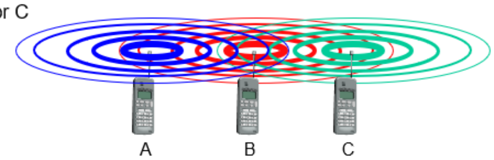
(-1)*(+1)* + *(-39)*(+1) + 19*(-1) + (-1)*(+1)* + *21*(-1) + (-39)*(+1) = -1 - 39 - 19 - 1 - 21 - 39 = -120

Step 10: Both results are negative. The receiver cannot reconstruct the original data of A, but it can still detect B's data because B's signal is stronger.

1. Hidden and Exposed terminals.

Hidden terminals

- A sends to B, C cannot receive A
- C wants to send to B, C senses a "free" medium (CS fails)
- collision at B, A cannot receive the collision (CD fails)
- A is "hidden" for C



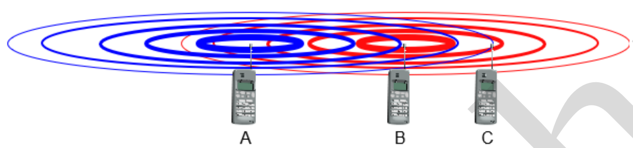
Exposed terminals

- B sends to A, C wants to send to another terminal (not A or B)
- C has to wait, CS signals a medium in use
- but A is outside the radio range of C, therefore waiting is not necessary
- C is "exposed" to B

❖ Near and far terminals:

Terminals A and B send, C receives

- signal strength decreases proportional to the square of the distance
- the signal of terminal B therefore drowns out A's signal
- C cannot receive A



If C for example was an arbiter for sending rights, terminal B would drown out terminal A already on the physical layer

Also severe problem for CDMA-networks - precise power control needed!

❖ What happens if a receiver has the wrong key or is not synchronized with the chipping sequence of the transmitter in CDMA? Describe with example.

In **CDMA**, each transmitter uses a unique **chipping sequence** (spreading code) to send data.

- **Wrong key:** If the receiver uses the wrong code, despreading fails → the output looks like random noise, and the original data is lost.
- **Not synchronized:** Even with the correct code, if timing is off, the data cannot be recovered → signal appears distorted.

Example:

- Transmitter sends bit 1 with code 1011.
- Correct receiver (code 1011) → gets 1.
- Wrong receiver (code 1100) → gets random data → cannot decode.

- **Suppose two senders, A and B, want to send data at the same time using the same frequency. CDMA assigns the unique key sequences: key A_k = 010011 for sender A, key B_k = 110101 for sender B.**

i. **Are the keys orthogonal? Prove.**

Given:

- Sender A key: 010011 → [-1, +1, -1, -1, +1, +1]
- Sender B key: 110101 → [+1, +1, -1, +1, -1, +1]
- Data: A = 1 (+1), B = 0 (-1)

Are keys orthogonal?

- Dot product = (-1)(+1)+(+1)(+1)+(-1)(-1)+(-1)(+1)+(+1)(-1)+(+1)(+1) = 0
- Yes, keys are orthogonal.

ii. **Illustrate the spreading and despreading of the data when sender A wants to send the bit $A_d = 1$, sender B sends $B_d = 0$. (code a binary 0 as -1, a binary 1 as +1 to apply the standard addition and multiplication rules)**

Spreading & despreading:

- Spread signals:
  - A: +1 × [-1, +1, -1, -1, +1, +1] = [-1, +1, -1, -1, +1, +1]
  - B: -1 × [+1, +1, -1, +1, -1, +1] = [-1, -1, +1, -1, +1, -1]
- Combined signal: [-2, 0, 0, -2, 2, 0]
- Despread at A: [-2,0,0,-2,2,0] × [-1,+1,-1,-1,+1,+1] → sum = 6 → bit = 1
- Despread at B: [-2,0,0,-2,2,0] × [+1,+1,-1,+1,-1,+1] → sum = -6 → bit = 0

Result: Each receiver recovers its correct bit.

- **Illustrate the components of an infrastructure and a wireless part as specified for IEEE 802.11. Explain each component shortly.**

1. Infrastructure Components:

- **Access Point (AP):** Connects wireless devices to the wired network and manages communication.
- **Distribution System (DS):** Connects multiple APs for wider coverage and roaming.
- **Basic Service Set (BSS):** Group of devices connected to one AP.
- **Extended Service Set (ESS):** Multiple BSSs connected via DS for larger network area.

2. Wireless Components:

- **Stations (STAs):** Wireless devices like laptops, phones, IoT devices.
- **Wireless Medium:** Air (radio waves) used for communication.
- **Network Interface Card (NIC):** Hardware in each device to send/receive wireless signals.